

Predicting stress using wearable sensors (University trial)

Registration number: 22014104

Link to GitHub: <https://github.com/Hemanth-Kanchibail-Bhat/University-Trail.git>

Executive summary (max. 200 words)	167
Main Findings and Discussion (max. 600 words)	595
Conclusions (max. 300 words)	300
Total word count	1062

Contents

1. Main Findings.....	2
1.1 Imbalance in label in the dataset	2
1.2 Check for normality distribution.....	3
1.3 Time series representation of the data	4
1.4 Correlation.....	5
1.5 Model Building and Evaluation	5
2. Discussion.....	8
3. Conclusion	9

Abstract

The goal of this study was to determine whether it was possible to forecast stress levels using wearable sensors in a company-developed watch. In pilot research that was carried out with the assistance of a university, the data set was examined using machine learning models such as Random Forest, Logistic Regression, GaussianNB, and Ada Boost. The company's goal was to reduce false negatives while creating an innovation that alerts customers when they are experiencing stress. According to the results, the Ada Boost model was the most accurate in predicting stress levels. The investigation revealed that key indicators for spotting stressful moments were measurements from the accelerometer sensor, heart rate, and electrodermal activity. The study concluded that the available wearable sensors can be used to detect and forecast stress, with the Ada Boost model recommended for implementation in the company's watch to enhance stress monitoring capabilities. Further research and evaluation of the algorithm's decision-making process, as well as the suitability of other signals, are suggested for future improvements.

1. Main Findings

1.1 Imbalance in label in the dataset

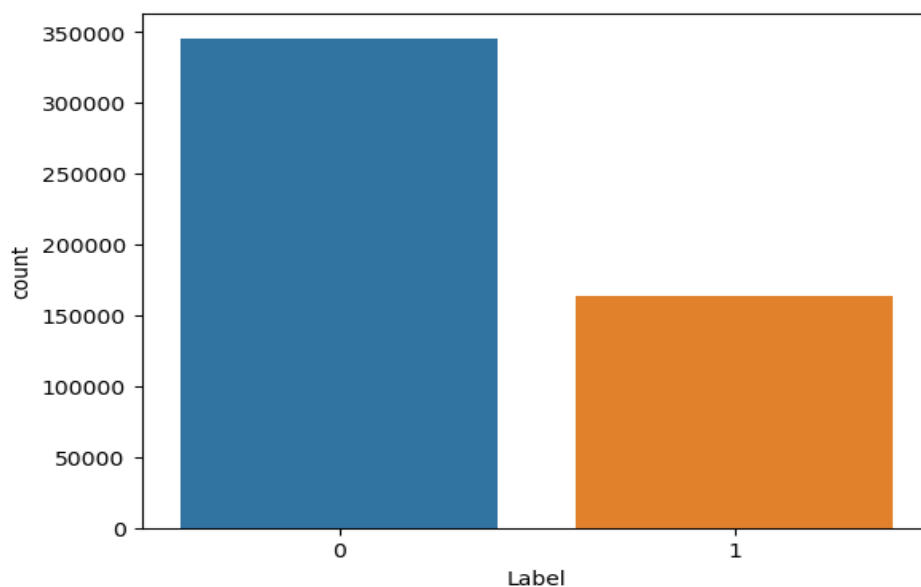


Figure-1: Imbalance of target variable in the dataset

From Figure-1, we can observe the imbalance in the target variable in the dataset. The imbalance in the label (Target variable) of the dataset leads to the resampling of the 0 and 1 ratio in the dataset. I performed manual balancing of data by randomly selecting samples from

the majority class (class 0) to match the number of samples in the minority class (class 1) in a dataset. This way the is retained and loss of data is prevented.

1.2 Check for normality distribution

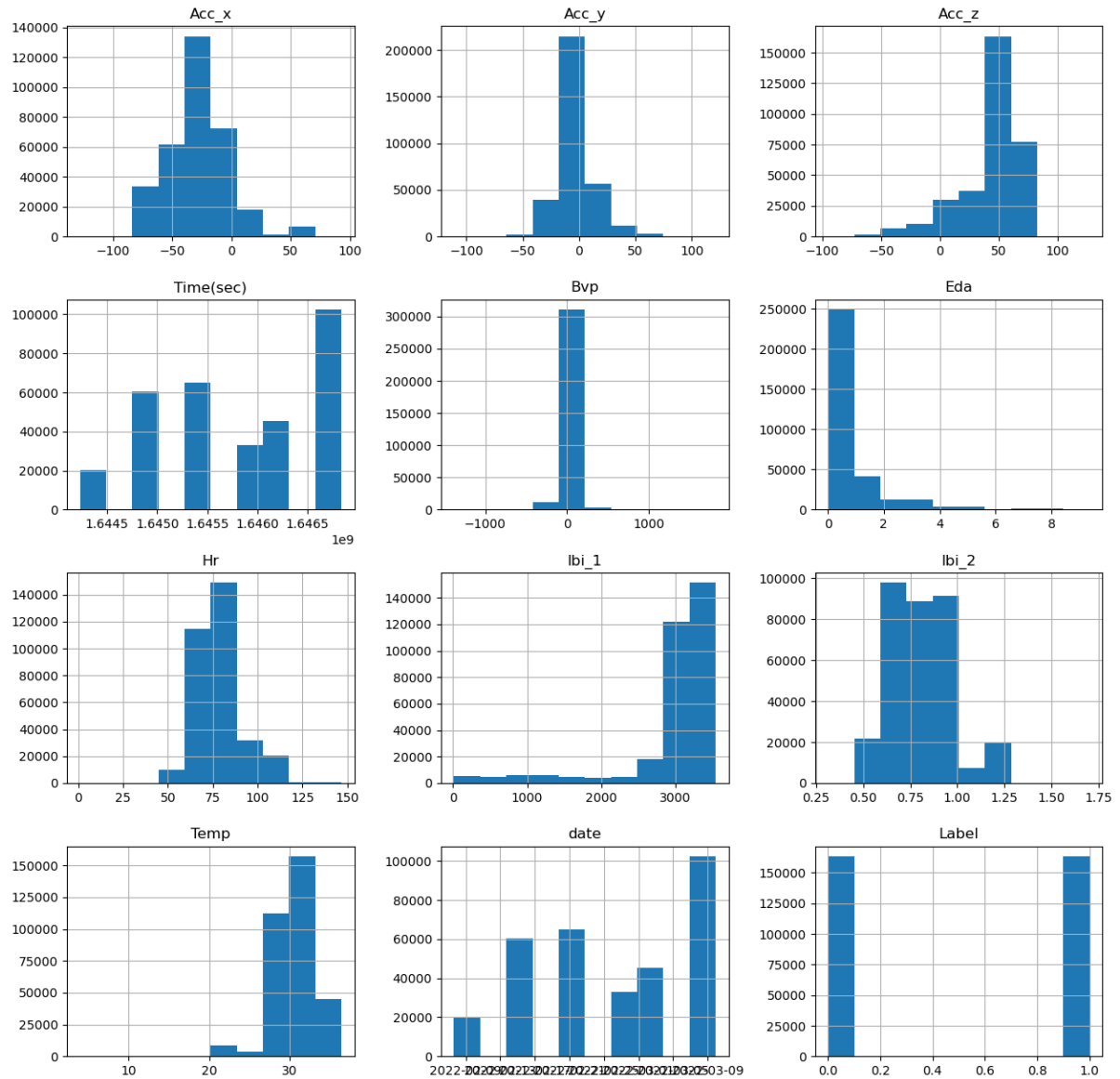


Figure-2: Histogram to check the Normality distribution of the data.

A histogram is a graphical representation of data that shows the distribution of values within a dataset. It is commonly used to visualize the frequency or count of data points falling into different ranges or bins. By examining the histogram of a dataset, we can visually assess if the data follows a normal distribution. If the histogram shows a symmetrical bell-shaped curve with no skewness, it may indicate a normal distribution. However, if the histogram displays significant asymmetry or lacks a bell-shaped curve, it may suggest that the data

does not follow a normal distribution. By looking at the above plot (Figure-2), it can be determined that data is not normally distributed.

1.3 Time series representation of the data

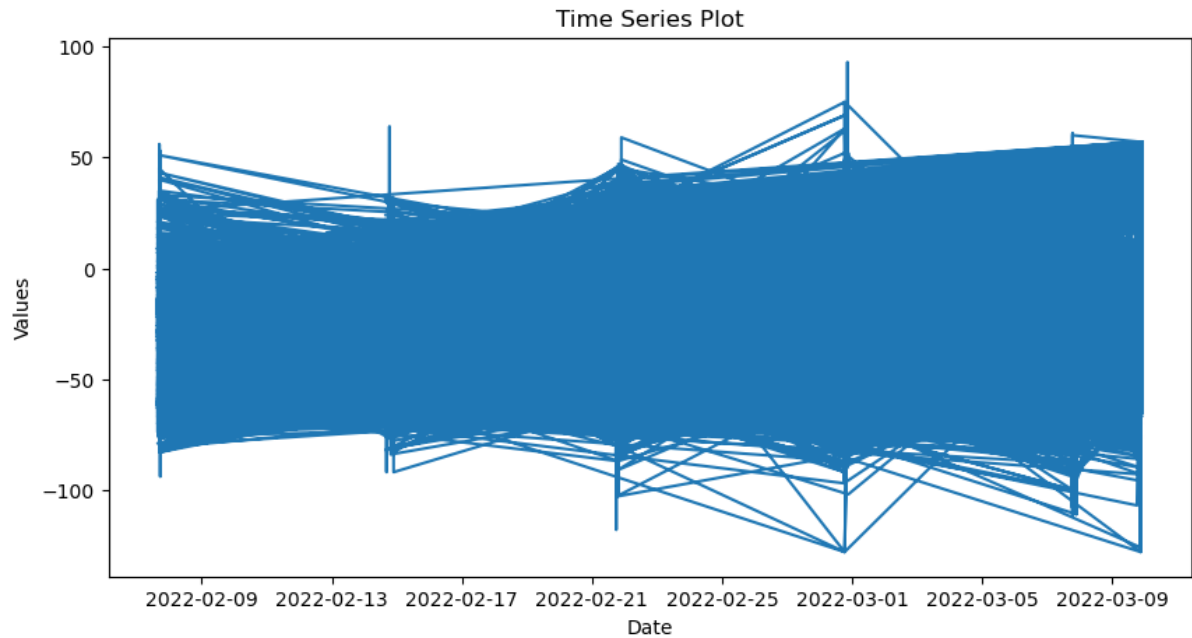


Figure-3: Time series representation of Accelerometer sensor reading.

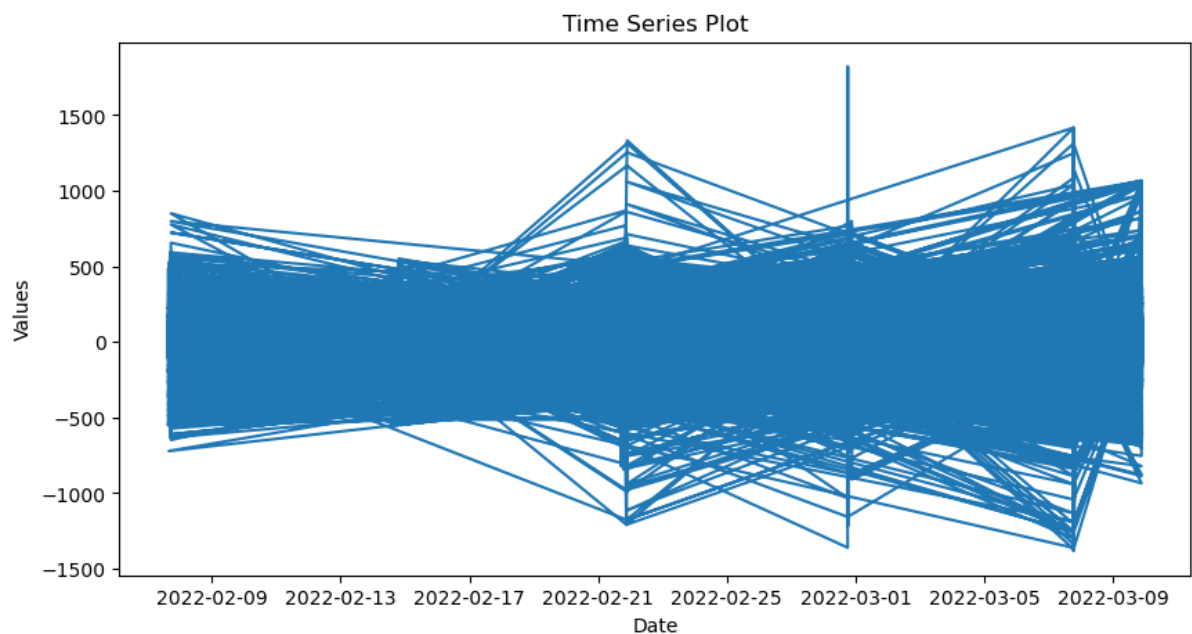


Figure-4: Time series representation of Blood Volume Pressure sensor reading

The above plots (Figure-3 and Figure-4) represent the change in values of the Accelerometer and Blood Volume pulse data with respect to the date it was recorded. This

shows when the candidates were experiencing stress at the most. From the above plots, the readings are high when candidates are experiencing stress.

1.4 Correlation

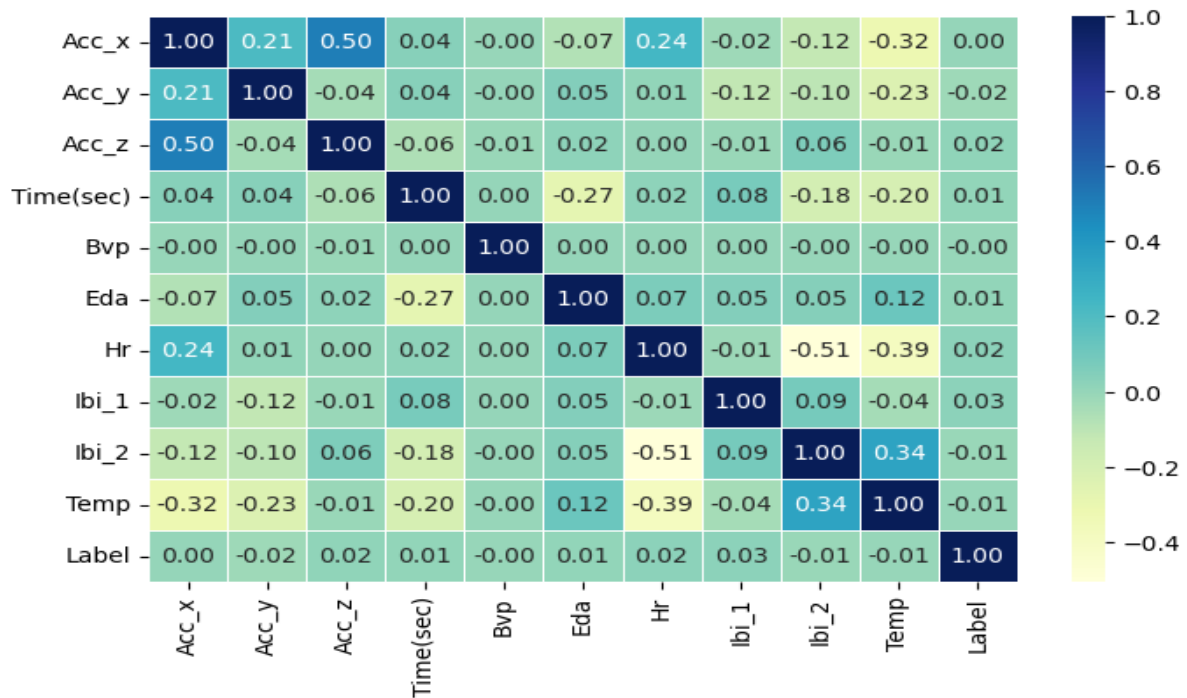


Figure-5: Correlation of variables in the dataset.

From the above correlation heatmap (Figure-5), determining stress is highly correlated with Acc_z, Eda, Hr, and lbi_2. It shows that the acceleration of motion of the body's z-axis along with Electrodermal Activity and Heart-rate are the major factors of stress.

1.5 Model Building and Evaluation

Upon completion of Data Pre-processing, The data is split into training and testing datasets having a test size of 20% of the original dataset. As this is a classification problem (Identifying if the candidate is undergoing stress or not), Four classification machine learning models were implemented. The Random Forest Classifier model, Logistic regression model, Gaussian Naïve Bayes model, and Ada Boost model were trained with the training data and tested on the testing

data to evaluate their performance on unexplored/new datasets. The Ada Boost model gave the best performance.

Ada Boost Model

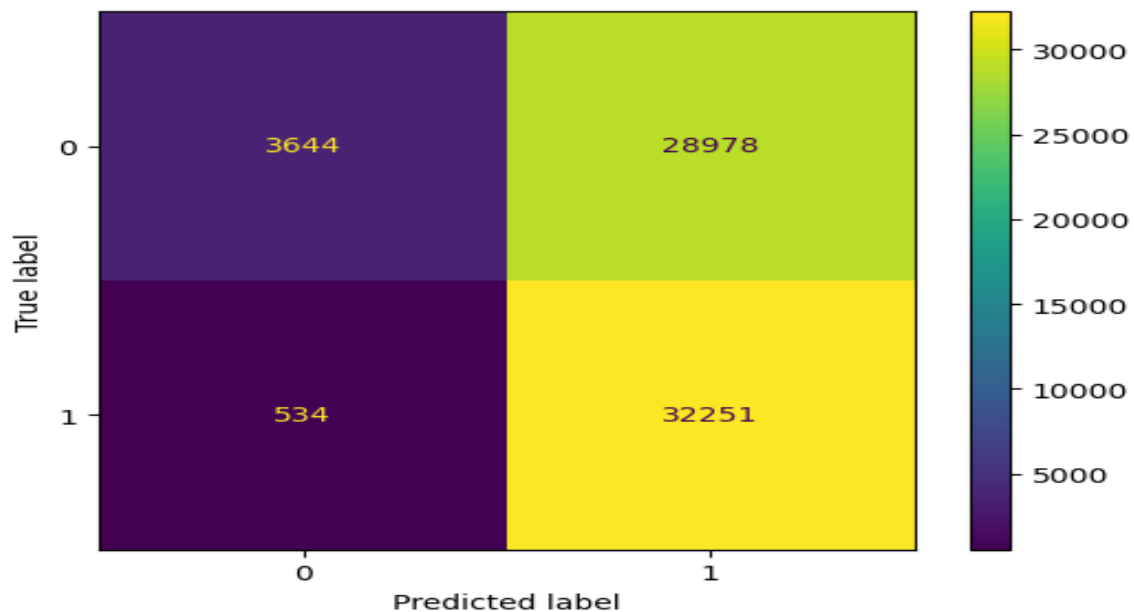


Figure-6: Confusion matrix of the Ada Boost Model.

Performance evaluation of the Ada Boost model:-

- Precision: 0.53
- Recall: 0.98
- F1-score: 0.69
- Accuracy: 0.55

From Figure-6 we can see that,

- 3644 represents the number of True Positives (TP), which are the instances that were actually positive (belonging to the positive class) and were predicted correctly as positive by the model.
- 28978 represents the number of True Negatives (TN), which are the instances that were actually negative (belonging to the negative class) and were predicted correctly as negative by the model.
- 534 represents the number of False Positives (FP), which are the instances that were actually negative but were predicted incorrectly as positive by the model.

- 32251 represents the number of False Negatives (FN), which are the instances that were actually positive but were predicted incorrectly as negative by the model.

Random Forest Classifier model

The next best-performing model is the Random Forest classifier. Figure-7 shows the confusion matrix performance evaluated by the Random forest classifier.

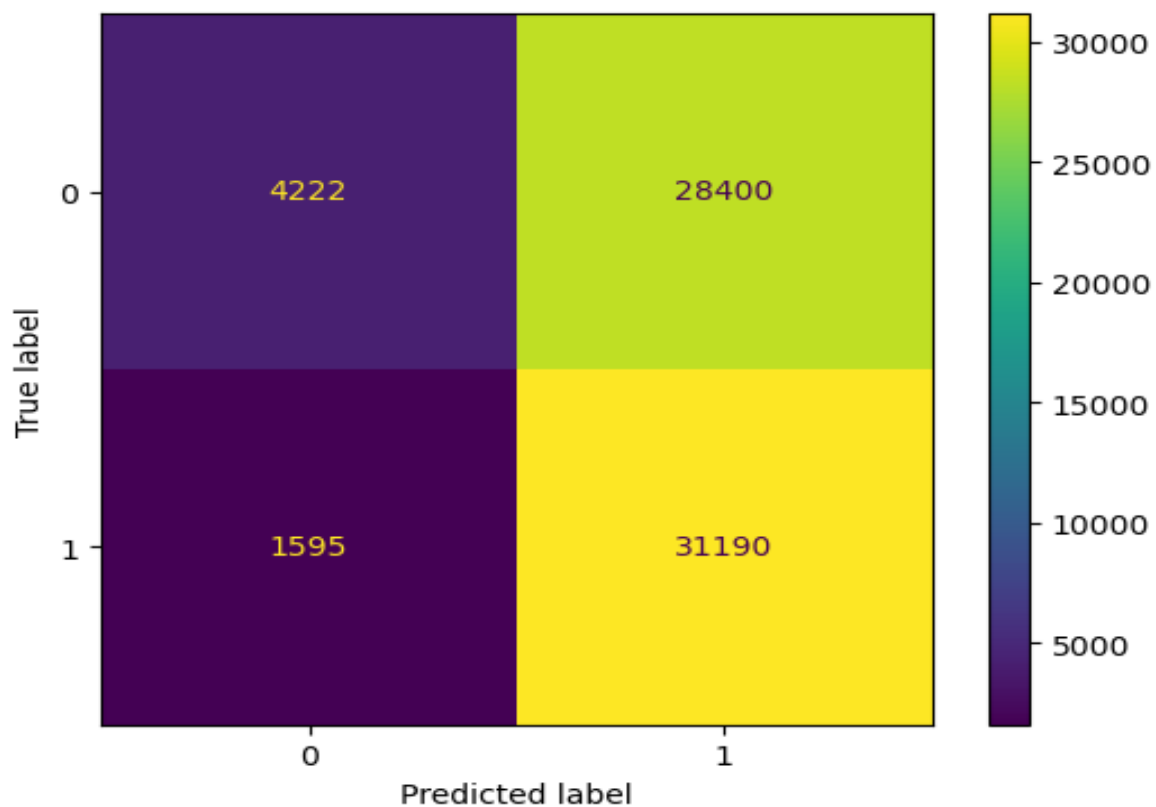


Figure-7: Confusion matrix of the Random Forest Classifier Model.

Performance evaluation of the Random Forest Classifier model:-

- Precision: 0.52
- Recall: 0.95
- F1-score: 0.68
- Accuracy: 0.54

From Figure-7 we can see that,

- 4222 represents the number of True Positives (TP), which are the instances that were actually positive (belonging to the positive class) and were predicted correctly as positive by the model.

- 28400 represents the number of True Negatives (TN), which are the instances that were actually negative (belonging to the negative class) and were predicted correctly as negative by the model.
- 1595 represents the number of False Positives (FP), which are the instances that were actually negative but were predicted incorrectly as positive by the model.
- 31190 represents the number of False Negatives (FN), which are the instances that were actually positive but were predicted incorrectly as negative by the model.

Feature Importance of each variable evaluated by the Random Forest Classifier model:

- Feature 'Acc_x': 0.017676453221082463
- Feature 'Acc_y': 0.0646399602986702
- Feature 'Acc_z': 0.10951880962607234
- Feature 'Time(sec)': 0.399771587575664
- Feature 'Bvp': 0.0030219541899028065
- Feature 'Eda': 0.020069818177957086
- Feature 'Hr': 0.02839672131820462
- Feature 'Ibi_1': 0.2855793203296448
- Feature 'Ibi_2': 0.005978424709186398
- Feature 'Temp': 0.06534695055361536

2. Discussion

The Stress-Predict dataset from the GitHub repository (<https://github.com/italha-d/Stress-Predict-Dataset>) was downloaded and accessed using `glob.glob()` which is used to search for files in the directory and returns a list of file paths that match the specified pattern, which are the folders containing the data for 35 volunteers. Forward fill (fill) and Backward fill (bill) methods to replace missing values with the previous and subsequent values, respectively, using the `fillna()` method. We discussed various performance metrics, including precision, recall, and F1-score, for assessing classification models. We also examined confusion matrices, which provide insights into the true positives, true negatives, false positives, and false negatives of a model's predictions. Among the models compared, the AdaBoost model stood out with the best performance, boasting the highest accuracy, precision, recall, and F1 score. We ascribed this result to AdaBoost's ensemble learning technique, which combines weak learners to generate a powerful classifier. This technique allows the model to handle complex datasets and improve

prediction accuracy by iteratively adjusting the weights of misclassified instances. As a result, AdaBoost is well-suited for classification tasks due to its ability to handle imbalanced data, reduce bias, and enhance overall model performance.

3. Conclusion

In conclusion, numerous important conclusions emerged from our investigation of stress categorization using machine learning models. On a dataset with physiological features like accelerometer data, electrodermal activity, heart rate, inter-beat interval, blood volume pulse, and temperature, we assessed the performance of four well-known classification algorithms, namely Logistic Regression, Gaussian Naive Bayes, Random Forest, and AdaBoost.

We found that the AdaBoost model outperformed the other three models in terms of accuracy, precision, recall, and F1-score, indicating its superior performance in accurately classifying stress levels. The AdaBoost model's ability to combine weak classifiers and adaptively re-weight samples during training, thus giving more importance to misclassified samples, may have contributed to its better performance. Furthermore, we observed that features such as 'Time(sec)', 'Ibi_1', 'Acc_z', and 'Temp' were found to be more important in identifying stress, based on the feature importance results. These features capture physiological changes associated with stress, such as time-dependent patterns, heart rate variability, body movement, and temperature changes.

Our findings highlight the importance of feature selection and model choice in stress classification tasks. By considering relevant physiological features and selecting appropriate machine learning models, we can improve the accuracy and effectiveness of stress detection systems. These findings have potential implications in various fields, such as healthcare, occupational safety, and mental health, where early detection and management of stress can be crucial.

However, it's important to note that our analysis has limitations, such as the use of a specific dataset and the potential influence of confounding factors. Further research with larger and more diverse datasets, including validation in real-world settings, is needed to validate our findings and improve the generalizability of the results.

In summary, our analysis suggests that the AdaBoost model, with consideration of important physiological features, can be effective in accurately classifying stress levels. These findings

contribute to the growing body of research on stress detection and management, and can potentially inform the development of practical applications in various domains to improve individual health and well-being.